Hands-On-Activity No. 5.2

Hands-on Activity 5.2: Structures

Course Code: CPE 007	**Program: Computer Engineering**
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1. Objective(s)

This activity aims to develop programming skills using structures.

2. Intended Learning Outcomes

2.1 Create and use structures

2.2 Pass structures to functions call by value and call by reference

3. Materials and Equipment

Internet connection
Personal Computer

4. Output

1. **Create a program that uses a structure to store a rectangle's length and width. Write a function that accepts the structure as an argument and computes the area and perimeter of the rectangle.**

Answer:

```
#include <iostream>

using namespace std;


struct Rectangle {

    double length;

    double width;

};


void calculateRectangle(const Rectangle& rectangle) {

    double area = rectangle.length * rectangle.width;

    double perimeter = 2 * (rectangle.length + rectangle.width);


    cout << "Area: " << area << '\n';

    cout << "Perimeter: " << perimeter << '\n';

}
```

```
int main() {
    Rectangle rectangle;

    cout << "Enter the rectangle's length: ";
    cin >> rectangle.length;

    cout << "Enter the rectangle's width: ";
    cin >> rectangle.width;

    calculateRectangle(rectangle);

    return 0;
}
```

ScreenShot:



```
#include <iostream>
using namespace std;

struct Rectangle {
    double length;
    double width;
};

void calculateRectangle(const Rectangle& rectangle) {
    double area = rectangle.length * rectangle.width;
    double perimeter = 2 * (rectangle.length + rectangle.width);

    cout << "Area: " << area << '\n';
    cout << "Perimeter: " << perimeter << '\n';
}

int main() {
    Rectangle rectangle;

    cout << "Enter the rectangle's length: ";
    cin >> rectangle.length;

    cout << "Enter the rectangle's width: ";
    cin >> rectangle.width;

    calculateRectangle(rectangle);

    return 0;
```

```
Enter the rectangle's length: 6
Enter the rectangle's width: 3
Area: 18
Perimeter: 18
```

2. **Write a program that creates a function multiple that determines if the integer entered from a keyboard is a multiple of some integer x.**

Answer:

```
#include <iostream>
```

```
using namespace std;

struct NumberPair {
    int number;
    int x;

};

bool multiple(const NumberPair& values) {
    return values.x != 0 && values.number % values.x == 0;
}

int main() {
    NumberPair values;

    cout << "Enter an integer: ";
    cin >> values.number;

    cout << "Enter the integer x: ";
    cin >> values.x;

    if (multiple(values)) {
        cout << values.number << " is a multiple of " << values.x <<
".\n";
    } else if (values.x == 0) {
        cout << "A number cannot be tested as a multiple of zero.\n";
    } else {
        cout << values.number << " is not a multiple of " << values.x
<< ".\n";
    }

    return 0;
}
```

ScreenShot:

```cpp
#include <iostream>

using namespace std;

struct NumberPair {
    int number;
    int x;

};

bool multiple(const NumberPair& values) {
    return values.x != 0 && values.number % values.x == 0;
}

int main() {
    NumberPair values;

    cout << "Enter an integer: ";
    cin >> values.number;

    cout << "Enter the integer x: ";
    cin >> values.x;

    if (multiple(values)) {
        cout << values.number << " is a multiple of " << values.x << ".\n";
    } else if (values.x == 0) {
        cout << "A number cannot be tested as a multiple of zero.\n";
    } else {
```

```
Enter an integer: 4
Enter the integer x: 5
4 is not a multiple of 5.
```

5. Conclusion

Learned about writing a Data String Structure for a more organized code.

6. Rubric

SO7